Unit 2 Chemical Reactions Test - Student Version (verK)

Subject: chemistry
Course Code: SCH3U
Unit: Unit 2 Chemical Reactions Test

Instructions: Answer all questions. Show full work where required. Write clearly and use appropriate notation.

Multiple Choice

Section Plan: 25 questions | 25 marks

Instructions: Choose the best answer for each question.

1. Which products are formed when calcium carbonate reacts with hydrochloric acid? Focus on the chemically best-supported answer rather than the most familiar one. (1 mark)

A. $CaO + HCl$
B. $CaCl_2 + H_2$
C. $Ca + Cl_2 + H_2O$
D. $CaCl_2 + H_2O + CO_2$

2. Which piece of evidence would best show that an acid spill has been successfully neutralized? Choose the best answer. Consider balancing, classification, and any evidence in the scenario before choosing. (1 mark)

A. The container is larger than before
B. The solution remains colorless only
C. The pH is near neutral after treatment
D. The reactants are still listed separately

3. Which balanced equation represents the complete combustion of propane? Select the most accurate option. Focus on the chemically best-supported answer rather than the most familiar one. (1 mark)

A. $C_3H_8 + 3O_2 \rightarrow 3CO + 4H_2O$
B. $C_3H_8 + 4O_2 \rightarrow 3CO_2 + 4H_2O$
C. $C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$
D. $2C_3H_8 + 7O_2 \rightarrow 6CO_2 + 8H_2O$

4. Why is notation precision especially important when comparing two possible product predictions for the same reactants? Consider balancing, classification, and any evidence in the scenario before choosing. (1 mark)

A. State symbols matter only in physics labs
B. Correct notation removes the need to balance

C. Notation only affects the appearance of the answer

D. Small notation differences can distinguish entirely different substances or states

5. Which pair of aqueous solutions is most likely to produce a precipitate? Choose the best answer. Focus on the chemically best-supported answer rather than the most familiar one. (1 mark)

A. HCl + H2O

B. KOH + NaOH

C. NaNO3 + KCl

D. BaCl2 + Na2SO4

6. Why is it important to include state symbols in a chemical equation when required? Select the most accurate option. Consider balancing, classification, and any evidence in the scenario before choosing. (1 mark)

A. They are decorative only

B. They replace coefficients

C. They determine atomic mass

D. They show the physical state of each substance

7. Which reaction is an example of decomposition? Focus on the chemically best-supported answer rather than the most familiar one. (1 mark)

A. 2H2O2 -> 2H2O + O2

B. 2Na + Cl2 -> 2NaCl

C. Mg + 2HCl -> MgCl2 + H2

D. NaOH + HCl -> NaCl + H2O

8. What does a blue Bunsen burner flame usually indicate compared with an orange flame? Choose the best answer. Consider balancing, classification, and any evidence in the scenario before choosing. (1 mark)

A. Water is the only product

B. No combustion is occurring

C. Less oxygen and cooler burning

D. More oxygen and more complete combustion

9. Which statement best describes incomplete combustion? Select the most accurate option. Focus on the chemically best-supported answer rather than the most familiar one. (1 mark)

A. It is the same as neutralization

B. It occurs when oxygen is abundant

C. It can produce carbon monoxide and soot

D. It always produces only carbon dioxide and water

10. Why can missing state symbols still cost marks even if the products are correct? Focus on the chemically best-supported answer rather than the most familiar one. (1 mark)

A. They are only needed in physics
B. State symbols replace balancing
C. They change the identity of all elements
D. They are required chemical notation and communicate important information

11. Which reactant pair is most likely to undergo an acid-carbonate reaction? Choose the best answer. Consider balancing, classification, and any evidence in the scenario before choosing. (1 mark)

A. O_2 + Ne
B. Ag + Au
C. HCl + $CaCO_3$
D. NaCl + KNO_3

12. Which observation best suggests that a precipitate has formed? Select the most accurate option. Focus on the chemically best-supported answer rather than the most familiar one. (1 mark)

A. The mass of each atom changes
B. The pH always becomes exactly 7
C. The solution becomes a gas only
D. A solid appears in a previously clear mixture

13. A student predicts products for a reaction between aluminum sulfate and calcium hydroxide. Which outcome is most chemically reasonable after balancing and considering solubility? Consider balancing, classification, and any evidence in the scenario before choosing. (1 mark)

A. $Al(OH)_3(s)$ + $CaSO_4(aq)$
B. $2Al(OH)_3(s)$ + $3CaSO_4(aq)$
C. Al(s) + $CaSO_4(aq)$ + $OH^-(aq)$
D. $Al_2O_3(s)$ + $CaSO_4(aq)$ + $H_2O(l)$

14. Why might a teacher require full sentences on application questions in chemistry? Choose the best answer. Focus on the chemically best-supported answer rather than the most familiar one. (1 mark)

A. To make balancing unnecessary
B. To avoid using equations entirely
C. To reduce the need for chemistry vocabulary
D. To show reasoning and communicate chemical ideas clearly

15. Which option best explains why the same hydrocarbon can produce different carbon-containing products under complete versus incomplete combustion? Select the most accurate option. Consider balancing, classification, and any evidence in the scenario before choosing. (1 mark)

A. State symbols determine the final products
B. Hydrogen stops reacting when temperature rises
C. Carbon changes atomic number during the reaction
D. Oxygen availability changes how fully carbon is oxidized

16. Which combination of evidence would be strongest for confirming that a clear acid spill has been neutralized rather than merely diluted? Focus on the chemically best-supported answer rather than the most familiar one. (1 mark)

A. The beaker feels cool and no odor is noticed

B. The solution remains transparent and unstirred

C. The solution is colorless and the volume increased

D. The pH is near neutral and reaction evidence is supported by a justified indicator or probe test

17. Which reaction below is best classified as a single displacement reaction? Choose the best answer. Consider balancing, classification, and any evidence in the scenario before choosing. (1 mark)

A. $2H_2O \rightarrow 2H_2 + O_2$

B. $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$

C. $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$

D. $AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$

18. A burner flame becomes luminous orange during heating. Which explanation best links the observation to reaction conditions and likely products? Select the most accurate option. Focus on the chemically best-supported answer rather than the most familiar one. (1 mark)

A. More oxygen is entering, so only water is produced

B. Combustion has stopped completely, so no products form

C. The gas is condensing, so a precipitation reaction begins

D. Less oxygen is entering, so incomplete combustion becomes more likely

19. Which pair of reactants is most likely to undergo a double displacement reaction? Consider balancing, classification, and any evidence in the scenario before choosing. (1 mark)

A. $Mg + O_2$

B. $CH_4 + O_2$

C. $Na + Cl_2$

D. $AgNO_3 + NaCl$

20. Which statement best explains why clear notation matters in chemistry? Choose the best answer. Consider balancing, classification, and any evidence in the scenario before choosing. (1 mark)

A. It reduces the need for observations

B. It changes the outcome of the reaction

C. It removes all experimental uncertainty

D. It helps communicate substances and reactions accurately

21. Which products are formed when sodium oxide reacts with sulfur trioxide? Select the most accurate option. Focus on the chemically best-supported answer rather than the most familiar one. (1 mark)

A. Na_2SO_4

B. $Na + SO_4$

C. $Na_2S + O_3$

D. $NaSO_3 + O_2$

22. Which result would best support the claim that combustion became more complete after adjusting a burner? Consider balancing, classification, and any evidence in the scenario before choosing. (1 mark)

A. The beaker disappears
B. The reactants stop reacting
C. The flame becomes cooler and more orange
D. The flame becomes bluer and less luminous

23. Which balanced equation best represents a single displacement reaction that is actually likely to occur based on the activity of the metals involved? Choose the best answer. Focus on the chemically best-supported answer rather than the most familiar one. (1 mark)

A. $Au + HCl \rightarrow AuCl_3 + H_2$
B. $Ag + NaCl \rightarrow AgCl + Na$
C. $Cu + MgSO_4 \rightarrow CuSO_4 + Mg$
D. $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$

24. Which test result would best show that a neutralized spill is now safe to clean up? Select the most accurate option. Consider balancing, classification, and any evidence in the scenario before choosing. (1 mark)

A. The pH is close to 7
B. The container feels heavier
C. The liquid is darker in color
D. The reactants are still separate

25. Which reaction below is best classified as a synthesis reaction? Focus on the chemically best-supported answer rather than the most familiar one. (1 mark)

A. $CaO + H_2O \rightarrow Ca(OH)_2$
B. $2KClO_3 \rightarrow 2KCl + 3O_2$
C. $HCl + NaOH \rightarrow NaCl + H_2O$
D. $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$

Predicting Products and Classifying Reactions

Section Plan: 8 questions | 35 marks

Instructions: Predict the product(s), include states where appropriate, balance the equation, and identify the reaction type.

26. Predict the products, balance the equation, and classify the reaction:
$Al + O_2 \rightarrow$ Use appropriate notation in your response. Justify each major step or classification in your response. (3 marks)

27. Describe what evidence you would record to show a chemical reaction occurred even when the solution stays clear. Include enough chemistry reasoning to show why your answer is valid, not just what it is. (3 marks)

28. Write the balanced equation with states for the reaction between silver nitrate and sodium chloride. Show the essential steps clearly. Justify each major step or classification in your response. (3 marks)

29. Describe a short test plan that would distinguish between a clear solution containing unreacted acid and a clear solution that has already been neutralized. Use appropriate notation in your response. Include enough chemistry reasoning to show why your answer is valid, not just what it is. (3 marks)

30. Predict the products, include states, balance the equation, and classify the reaction:
$AgNO_3(aq) + MgBr_2(aq)$ -> Include enough chemistry reasoning to show why your answer is valid, not just what it is. (3 marks)

31. Describe one real-world application of combustion reactions and one safety concern associated with it. Show the essential steps clearly. Justify each major step or classification in your response. (3 marks)

32. Complete, balance, and classify the reaction:
$Na_2O + SO_3$ -> Use appropriate notation in your response. Include enough chemistry reasoning to show why your answer is valid, not just what it is. (3 marks)

33. A student writes the correct products for a reaction but omits all state symbols. Explain how this can reduce scientific quality and give one situation where the missing state would change the interpretation of the reaction. Justify each major step or classification in your response. (3 marks)

Application Questions

Section Plan: 5 questions | 18 marks

Instructions: Use full sentences for all questions to achieve full marks.

34. An antacid tablet containing calcium carbonate reacts with stomach acid. Explain why this is a neutralization-related process and include a balanced equation with states. Support your answer with clear reasoning. Connect your explanation to both reaction theory and real lab or real-world evidence. (4 marks)

35. Why is using correct chemical notation important when recording observations and equations in a chemistry lab? Include enough chemistry reasoning to show why your answer is valid, not just what it is. (3 marks)

36. A clear acid spill is neutralized and no visible change is observed. Describe how you could prove the reaction occurred and the area is safe to clean. Connect your explanation to both reaction theory and real lab or real-world evidence. (4 marks)

37. Explain why a blue Bunsen burner flame is preferred for heating in the lab. Use appropriate notation in your response. Include enough chemistry reasoning to show why your answer is valid, not just what it is. (3 marks)

38. During a lab, two groups obtain different products for the combustion of the same hydrocarbon sample. Explain how oxygen supply, flame appearance, and product testing could resolve which group observed complete combustion. Use a real-world explanation where appropriate. Connect your explanation to both reaction theory and real lab or real-world evidence. (4 marks)